

Math 215A. Fall 2024. Homework answers

HW1

1. Which of the following spaces are pairwise homeomorphic and which are not: (a) the orthogonal group SO_3 , (b) the space T_1S^2 of unit tangent vectors to S^2 , (c) the Stiefel manifold $V(3, 2)$, (d) in the complex 3-space $\mathbb{C}^3$ with coordinates z_1, z_2, z_3 , the set of unit vectors (i.e. $|z_1|^2 + |z_2|^2 + |z_3|^2 = 1$) satisfying $z_1^2 + z_2^2 + z_3^2 = 0$, (e) the real projective space $\mathbb{R}P^3$, (f) $S^2 \times S^1$?

Answer: All except (f) are homeomorphic to each other. In (a,b,c), each space consists of ordered pairs of perpendicular vectors of fixed lengths (in the SO_3 case, a matrix's third column is the cross-product of the first two). In (d), writing each z_i as $x_i + \sqrt{-1}y_i$ we obtain the relations $\sum x_i^2 + \sum y_i^2 = 1$, $\sum x_i^2 - \sum y_i^2 = 0$, and $2 \sum x_i y_i = 0$, which lead to the same description: a pair $(\vec{x}, \vec{y})$ of orthogonal vectors of length $1/\sqrt{2}$ each.

The identification of $\mathbb{R}P^3$ with SO_3 is famous but less straightforward. Consider the action of the group $Sp_1 = S^3$ of unit quaternions on $\mathbb{H}$ by conjugations: $x \mapsto uxu^{-1}$. It leaves the 3-dimensional space of $bi + cj + dk$ of purely imaginary quaternions invariant and acts on it by (length-preserving and hence) orthogonal transformations, which defines a homomorphism $Sp_1 \rightarrow SO_3$. It is not hard to see that it is surjective, with the kernel ± 1 . Thus, SO_3 is identified with $S^3/(\pm 1) = \mathbb{R}P^3$.

The fact that T_1S^2 and $S^2 \times S^1$ are *not* homeomorphic is equivalent to the famous Hairy Ball Theorem (non-existence of nonvanishing vector fields on the 2-sphere). At this point in the course we actually have no tools to prove it; so, if you tried and failed, it's fine: the better your attempt can serve as a motivation for developing the tools. For example, we will soon develop the theory of fundamental groups π_1 , and find that $\pi_1S^2 \times S^1 = \mathbb{Z}$ and $\pi_1\mathbb{R}P^3 = \mathbb{Z}_2$, which implies the statement.

2. Prove that a compact subset in $\mathbb{R}^\infty := \lim \mathbb{R}^n$ (the union of a nested sequence of Euclidean spaces) must lie in a finite dimensional subspace $\mathbb{R}^n$.

Answer. If not then the compact set contains a sequence — and hence a convergent subsequence x_n — with the property $x_n \notin \mathbb{R}^n$, and moreover, one may assume that the limit is none of x_n . Then the set $\{x_n\}$ is not closed in $\mathbb{R}^\infty$ in conflict with the fact that its intersection with every $\mathbb{R}^n$ is finite and hence closed.

I was criticized for saying that an infinite sequence in a compact space has a convergent subsequence — because in general spaces this might not be true. However, it is true that every infinite set in a (Hausdorff) compact

space has a limit point, and I should have said that my subset $\{x_n\}$ with each $x_n \notin \mathbb{R}^n$ doesn't, since every subset of it is closed.

3. Exercise 20 from Lecture 2: For a base-point space (X, x_0) construct a natural with respect to X homeomorphism between the suspension of X and the smash-product of X with S^1 .

Answer. Represent S^1 as the quotient of $[0, 1]$ by the endpoints. Then $X \# S^1$ is obtained from $X \times [0, 1]$ by contracting $X \times 0 \cup X \times 1 \cup x_0 \times [0, 1]$ — and so is $\Sigma(X, x_0)$. Moreover, a map $f : (X, x_0) \rightarrow (Y, y_0)$ defines $f \times \text{id}_{[0,1]} : X \times [0, 1] \rightarrow Y \times [0, 1]$ which maps the to-be-contracted subsets into each other, and hence descends to the quotients.

HW2

1. For the manifold of complete flags $V^1 \subset V^2 \subset \mathbb{C}^3$ in (say, complex) 3-space (actually the field doesn't matter), describe explicitly which flags belong to which cell (the total number should be 3!)

Answer. (a) Let $\mathbb{C}^1 \subset \mathbb{C}^2 \subset \mathbb{C}^3$ be the reference flag. The 0-dimensional cell is the flag with $V^1 = \mathbb{C}^1, V^2 = \mathbb{C}^2$. There are two 1-dimensional cells: one with $V_2 = \mathbb{C}^2$ and V^1 any line in $\mathbb{C}^2$ other than $\mathbb{C}^1$, the other with $V_1 = \mathbb{C}^1$ and V^2 any plane containing $\mathbb{C}^1$ other than $\mathbb{C}^2$. There are two 2-dimensional cells: one with $V^2 \neq \mathbb{C}^2$ containing $\mathbb{C}^1$ and V_1 any line in V^2 other than $\mathbb{C}^1$, the other with V_1 any line in $\mathbb{C}^2$ other than $\mathbb{C}^1$ and V_2 any plane containing that line other than $\mathbb{C}^2$. There is one 3-dimensional cell: with V_1 any line not contained in $\mathbb{C}^2$ and V_2 any plane containing V_1 and not containing $\mathbb{C}^1$.

2. Let c_k be the number of k -dimensional (in complex units) Bruhat cells in the manifold of complete flags in an n -dimensional complex space. Show that the generating function $c_0 + c_1q + c_2q^2 + \dots$ of this sequence is equal to the "q-factorial": $\prod_{k=1}^n (1 - q^k)/(1 - q)$, and check this for your answer in **1**.. *Hint:* Consider the Bruhat cell partition of the manifold of complete flags in the space F^n , where F is a finite field with q elements.

Answer. A complete flag in F^n is specified by one of $q^n - 1$ non-zero vectors $v \in F^n - 0$ modulo $q - 1$ rescalings by non-zero scalars, and by a complete flag in the quotient space $F^n / \text{span}(v)$. We conclude that the number f_n of complete flags in F^n is equal to $f_{n-1}(q^n - 1)/(q - 1)$, and by induction $f_n = \prod_{k=1}^n (q^k - 1)/(q - 1)$. On the other hand, each Bruhat cell of dimension k is parameterized by points of an affine k -dimensional space F^k which consists of q^k elements. If c_k is the number of such cells, then $f_n = \sum c_k q^k$ as required. In particular $f_3 = (q^3 - 1)(q^2 - 1)(q - 1)/(q - 1)^3 =$

$(1 + q + q^2)(1 + q) = 1 + 2q + 2q^2 + q^3$ in agreement with part (a).

3. Prove that S^∞ is contractible. *Hint:* Show first that S^n is contractible inside S^{n+1} .

Answer. Since the upper hemisphere with respect to the equatorial $S^n \subset S^{n+1}$ is topologically a disk D^{n+1} , the equatorial sphere can be deformationally retracted to the base point $(1, 0, \dots, 0)$ within the hemisphere. By Borsuk's theorem, this retraction can be extended to a homotopy $F^{(n)} : S^\infty \times I \rightarrow S^\infty$. Moreover, by the cellular approximation theorem, we may assume that $F^{(n)}$ is cellular, and hence at its last moment it maps $S^n = sk_n S^\infty$ to the base point $(1, 0, 0, \dots)$ and $S^{n+1} = sk_{n+1} S^\infty$ into itself. Next, we concatenate $F^{(n)}$ with $F^{(n+1)}$, i.e. *compose* each $F_t^{(n+1)}$ with $F_1^{(n)}$. The final map $F_1^{(n+1)} \circ F_1^{(n)}$ of this concatenation is cellular and contracts $S^{n+1} = sk_{n+1} S^\infty$ to the base point. Continuing this way and designating $[\frac{n-1}{n}, \frac{n}{n+1}]$ as the time interval for $F^{(n)}$, we obtain a continuous cellular map $F : S^\infty \times [0, 1] \rightarrow S^\infty$ which maps $S^n = sk_n S^\infty$ to the base point for all $t \geq \frac{n}{n+1}$. Thus, the constant map F_1 of S^∞ to the base point is the pointwise limit of F_t as $t \rightarrow 1^-$, and it extends F to $t = 1$ continuously due to the nature of cellular topology (the W-axiom). Thus extended, F provides a deformation retraction of S^∞ to the base point.

HW3

1. Prove that the fundamental group of a loop space ΩX of any base point space (X, x_0) is abelian. The same for any topological group. (*Hint:* Start with the group T^2 .)

Answer. Given two loops $\alpha, \beta : (S^1, 0) \rightarrow \Omega X$, use the operation $\circ$ of loop composition to define a map $S^1 \times S^1 \rightarrow \Omega X : (t, u) \mapsto \alpha(t) \circ \beta(u)$ of the torus. The map sends the generators $S^1 \times 0$ and $0 \times S^1$ of the 2-torus to α and β respectively. Since these generators commute in $\pi_1(S^1 \times S^1) = \mathbb{Z}^2$, their images in $\pi_1(\Omega X)$ also commute. Here ΩX can be replaced with any based point space (Y, y_0) equipped with a based point map $Y \times Y \xrightarrow{\circ} Y$ such that both $Y \times y_0 \xrightarrow{\circ} Y$ and $y_0 \times Y \xrightarrow{\circ} Y$ are id_Y , and with any topological group in particular.

2. Every non-orientable connected manifold has the *orienting* double-cover, whose fiber over a given point consists of the two orientations (of the tangent space) at that point. Find out which of the double-covers $G_+(n, k) \rightarrow G(n, k)$ are orienting.

Answer. All such $G(n, k)$ except $G(2, 1) = S^1$ have $\pi_1 = \mathbb{Z}_2$. This follows from the Schubert cell structure, with $sk_1 = G(2, 1)$ and sk_2 consisting of $G(3, 1) = \mathbb{R}P^2$ or $G(3, 2) = \mathbb{R}P^2$ or their union. Therefore, with that exception, $G_+(n, k)$ is the universal cover, and is the orienting one if and only if $G(n, k)$ is not orientable, and we claim that this is the case exactly when n is odd. Indeed, the tangent space to a grassmannian $G(n, k)$ at a point $V^k \subset \mathbb{R}^n$ is canonically identified with $\text{Hom}(V^k, \mathbb{R}^n/V^k) \cong V^* \otimes (\mathbb{R}^n/V^k)$. If $\{e_i\}$ and $\{e'_j\}$ are bases in V and $\mathbb{R}^n/V$ respectively, then $e_i^* \otimes e'_j$ form a basis in the tensor product. To find out whether a change of orientations in V and $\mathbb{R}^n/V$ without changing the orientation of $\mathbb{R}^n$ (i.e. passing from one branch of the double cover $G_+(n, k) \rightarrow G(n, k)$ to the other) changes the orientation of $V^* \otimes (\mathbb{R}^n/V)$, replace e_1 with $-e_1$ and e'_1 with $-e'_1$. Then $e_1^* \otimes e'_j$ with $j \neq 1$ and $e_i^* \otimes e'_1$ with $i \neq 1$ changes the sign while all other $e_i^* \otimes e'_j$ remain unchanged. The determinant of this basis change in $V^* \otimes (\mathbb{R}^n/V)$ equals $(-1)^{n-k-1+k-1} = (-1)^n$. Thus, $G(n, k)$ ($k \neq 0, n$) is orientable if and only if n is even.

3. (a) Show that if X is locally path-connected, then the projection $E(X, x_0) \rightarrow X$ is *open*, i.e. the image of every open set is open). (b) If in addition X is semi-locally simply connected, then $E(X, x_0)$ is locally path-connected.

Answer. (a) Let $\gamma : [0, 1] \rightarrow X$ be a point in the image of an open subset, $\gamma(0) = x_0$, $\gamma(1) = x_1$. Inside that open set γ has a neighborhood defined by the conditions that certain compact subsets $K_1, \dots, K_n$ of $[0, 1]$ are mapped respectively to certain open subsets $U_1, \dots, U_n$ in X . Let U be the intersection of those U_i for which $K_i \ni 1$ (and hence $\gamma(1) \in U$). By the continuity of γ , some interval $[1 - \delta, 1]$ is mapped by γ into U . Moreover, the interval can be selected short enough that it is disjoint from all K_i not containing 1, and that $\gamma(1 - \delta)$ lands in a neighborhood V of $\gamma(1)$ satisfying property A: any two points of V can be connected by a path in U . Clearly a path $\gamma' : [0, 1] \rightarrow X$ which coincides with γ on $[0, 1 - \delta]$ and maps $[1 - \delta, 1]$ to a path in U ending at $\gamma'(1) \in V$ lie in a smaller neighborhood of γ defined by adding the last two “compact-open” conditions. The image of this neighborhood in X coincides with V (since $\gamma(1 - \delta)$ can be connected this way to any point in V). Thus, the image of an open set is open because every point of it is contained in the image together with a neighborhood of it. (b) By definition every point $x \in X$ has a neighborhood U satisfying property B: every loop in U is contractible in X . Inside U , we can pick

a smaller neighborhood V of x satisfying property A. Given $\gamma : [0, 1] \rightarrow X$ ($\gamma(0) = x_0$), by compactness of $[0, 1]$, for N large enough there exist neighborhoods U_k with property B such that $\gamma([(k-1)/N, k/N]) \subset U_k$ for each $k = 1, \dots, N$. Inside each $U_k \cap U_{k+1}$ pick a neighborhood $V_k \ni \gamma(k/N)$ satisfying property A (here $k = 1, \dots, N-1$). The set of all paths γ' satisfying the conditions: $\gamma'([(k-1)/N, k/N]) \subset U_k$ and $\gamma'(k/N) \in V_k$ form an open neighborhood $\mathcal{U}$ of γ in the compact-open topology of $E(X, x_0)$.¹ Inside U_N pick a neighborhood V_N of $\gamma(1)$ satisfying property A, and define $\mathcal{V} \subset \mathcal{U}$ by the additional condition $\gamma'(1) \in V_N$. Given $\gamma' \in \mathcal{V}$, connect $\gamma(k/N)$ with $\gamma'(k/N)$ by a path δ_k inside $U_k \cap U_{k+1}$ (inside U_N for $k = N$ and constant for $k = 0$). Represent the loop formed by $\gamma' \delta_N^{-1} \gamma^{-1}$ as the composition of small loops formed by $\delta_{k-1}, \gamma'|_{[(k-1)/N, k/N]}, \delta_k^{-1}, \gamma^{-1}|_{[(k-1)/N, k/N]}$. By construction, each small loop lies in U_k and hence is contractible in X . Consequently their composition $\gamma' \delta_N^{-1} \gamma^{-1}$ is contractible, and hence γ' is homotopic to γ inside $\mathcal{U}$.

HW4

1. Prove that the join of two spaces, of which at least one (at first each) is path-connected, is simply connected.

Answer. By Van Kampen's theorem $\pi_1(X * Y)$ of two path-connected spaces is the quotient of the free product $\pi_1(X)$ and $\pi_1(Y)$ by the normal subgroup generated by the relations which come from $\pi_1(X \times Y) = \pi_1(X) \oplus \pi_1(Y)$ which come in the form $x = (x, y) = y$ for any $x \in \pi_1(X)$, $y \in \pi_1(Y)$, implying $x = (x, e) = e$ and $e = (e, y) = y$. (The two open sets used here are $X * Y - Y$ and $X * Y - X$.)

When only Y is path-connected, the following argument (which I learned during students' discussion at my office hours) consists in using the same open

¹Moreover, such $\mathcal{U}$ form a base of neighborhoods of γ (because property B neighborhoods form a base of topology in X). Indeed, given a compact $K \subset [0, 1]$ and an open $W \subset X$ containing $\gamma(K)$, for each $t \in [0, 1]$ take a neighborhood U_t from the base (i.e. with property B in our case) of $\gamma(t)$ in X , such that it fits W whenever $t \in K$. Using continuity of γ , take for each t an interval $I_t \subset [0, 1]$ containing t , disjoint from K when $t \notin K$, and mapped by γ into U_t . The intervals form an open cover of $[0, 1]$ and hence for N sufficiently large each $[(k-1)/N, k/N]$ fits one of the intervals, I_{t_k} . Therefore all path γ' mapping $[(k-1)/N, k/N]$ to U_{t_k} form a neighborhood of γ inside the open set defined by K and W in the compact-open topology of the path space. If the initial neighborhood of γ in the path space is defined by several pairs (K_j, W_j) , still for N large enough all path γ' mapping $[(k-1)/N, k/N]$ to $\cap_j U_{t_k}^j$ form a neighborhood of γ in the path space contained in the neighborhood defined by the conditions $\gamma'(K_j) \subset W_j$.

sets and applying an argument from the proof of Van Kampen's theorem: partitioning the interval $[0, 1]$ parameterizing the loop into N parts each lying one of the two open sets. This allows one to replacing the loop with a composition of loops each lying in some $X_0 * Y$, where X_0 is one of path-connected components of X , and hence contractible by the first part.

Alternatively (and this was my original plan), one can do induction on the number of path-connected components of X intersected by a given loop, using the first part as the base. In this approach, we have space W obtained by gluing along Y : the simply-connected (by the induction hypothesis) join Z of Y with the disjoint union of all but one of the path-connected components of X , and $X_0 * Y$ where X_0 is the remaining component. Applying Van Kampen's theorem to $W = U_1 \cup U_2$ with $U_1 = W - Z \sim X_0$, $U_2 = W - X_0 \sim Z$ and $U_1 \cap U_2 = X_0 \times (0, 1) \times Y$ we conclude that as before that W is simply connected.

A difficulty with the second approach is that path-connected components of X might not be open in X . This can be fixed by means of weakening the topology by declaring them open in X , and noting that due to the path-connectedness of $[0, 1]$ a loop continuous in the original topology still remains continuous in the weakened one. In particular, this implies (due to the compactness of $[0, 1]$) that each loop intersects finitely many path-connected components of X , and is therefore homotopic to a loop lying in the join of Y with these finitely many components, thereby enabling the induction on their number.

2. Let X be the space obtained by attaching one end of a 2-dimensional cylinder to the other end by the double cover map of the circle. (This space is sometimes called the "mapping torus" of the double-cover map of S^1 to S^1 .) Compute all homotopy groups of X . *Hint:* First solve the problem for the regular $\mathbb{Z}$ -covering of X obtained by unwinding the mapping cylinder along the generator.

Answer. Describing the space as a bouquet of two circles a, b with a 2-cell attached, we find the presentation of the fundamental group π_1 by the relation $aba^{-1}b^{-2} = e$. Note that $b = (a^{-1}ba)^2 = (a^{-2}ba^2)^4 = \dots$, i.e. π_1 contains a normal subgroup isomorphic (in its additive notation) to binary rationals $\mathbb{Z}[1/2] = \{m/2^k\}$. The quotient by this subgroup (described in generators and relations by declaring $b = e$) is the infinite cyclic group generated by a . Thus, π is the semidirect product of $\mathbb{Z}$ with $\mathbb{Z}[1/2]$ corresponding to the action of a (resp. a^{-1} by conjugations on $\mathbb{Z}[1/2]$ as the multiplication (resp.

division) by 2.

The regular $\mathbb{Z}$ -covering of X corresponding to this normal subgroup is obtained by gluing together a two-sided infinite sequence of cylinders $S^1 \times I$ by attaching the right end of a previous cylinder to the left end of the next one via the double-cover map $S^1 \rightarrow S^1$. A spheroid (of dimension > 1) mapped to this covering space lands in a finite collection of such cylinders, which can be deformationally retracted to its rightmost end S^1 where the spheroid is contractible (since the universal cover of S^1 is). This, $\pi_n(X) = 0$ for $n > 1$.

3. Exercise 16 from 8.8: If one removes ϕ_3 from the 5-lemma diagram on page 102, will it still remain true that $A_3 \equiv B_3$?

Answer: No. Assuming that $A_1 = B_1 = A_5 = B_5 = 0$, we find that A_3 and B_3 have the same subgroup $A_2 = B_2$ and the same quotient group $A_4 = B_4$, which is true for two non-isomorphic groups $\mathbb{Z}_4$ and $\mathbb{Z}_2 \oplus \mathbb{Z}_2$.